Electroplating

Exam code: 0620 & 0971

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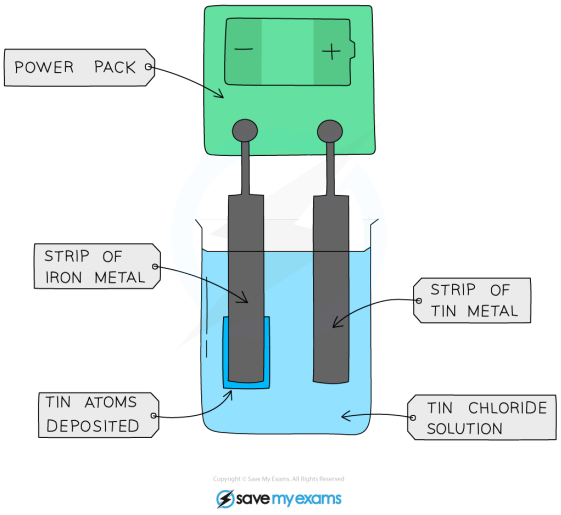
Electroplating

What is electroplating?

- Electroplating is a process where the surface of one metal is **coated** with a layer of a different metal
- The **cathode / negative electrode** is the **object** to be electroplated
- The **anode / positive electrode** is made from the **pure** metal that will be plated onto the object
- The **electrolyte** is an **aqueous solution** of a soluble salt of the pure metal at the anode

Electroplating a strip of iron with tin



A piece of iron being electroplated with tin. The electrolyte is tin(II) chloride, a water-soluble salt of tin

- At the anode:
  - Tin atoms lose electrons to form tin ions in solution
  - The loss of electrons is oxidation
- At the cathode:
  - Tin ions gain electrons to form tin atoms
  - The gain of electrons is reduction

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Examiner Tips and Tricks

Extended Tier students may be asked to write the ionic half equations for the reaction at each electrode. For the example above, these would be:

At the anode: $Sn\,(s) \rightarrow Sn^{2+}\,(aq) + 2e^-$

At the cathode: $Sn^{2+}\,(aq) + 2e^- \rightarrow Sn\,(s)$

What is electroplating used for?

- Electroplating is done to make metals more **resistant** to corrosion or damage
  - e.g. chromium and nickel plating
  - e.g. galvanising - zinc plating / coating, typically done to iron and steel
- It is also done to **improve the appearance** of metals,
  - e.g. coating cutlery and jewellery with silver

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